

Customer Journey Map

8.PROJECT DEMONSTRATION

Date	4 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	1 Mark

Scalability & Future Scope Plan

This document details the roadmap for expanding the feature set, modeling architecture, and deployment scalability of the HDI Predictor.

1. Feature Set Expansion

1. Gender Inequality Index (GII) Predictor:

- Integrate gender-disaggregated indicators (maternal mortality, adolescent birth rates, parliamentary seats share, educational attainment).
- Predict GI alongside HDI to offer a multi-dimensional picture of development equity.

4. Multidimensional Poverty Index (MPI) Calculator:

- Accept deprivation indicators across health (nutrition, child mortality), education (school enrollment), and standard of living (cooking fuel, sanitation, water, assets).
 - Classify severe poverty segments dynamically.
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2. Advanced Predictive Modeling

1. Non-Linear Models (Random Forests & Gradient Boosting):

2. While Linear Regression performs exceptionally well, random forest regressors can model complex joint feature distributions, reducing error offsets.

3. Time-Series Forecasting (ARIMA / LSTM):

4. Incorporate historic time-series data to predict country development paths (e.g. estimating a country's HDI score for 2030 based on current growth trends).

3. Cloud Integration & Scalability

1. SQL Database Migration:

2. Replace LocalStorage caching with a centralized relational database (e.g. **PostgreSQL** or **SQLite**) managed via **SQLAlchemy**.

3. Secure user accounts with database hashed passwords (bcrypt).

4. Dockerization:

5. Package the application in a lightweight **Docker** container to run consistently on AWS, GCP, or Azure:

```
dockerfile FROM python:3.11-slim WORKDIR /app COPY requirements.txt . RUN pip install --no-cache-dir -r requirements.txt COPY . . EXPOSE 5000 CMD ["gunicorn", "--bind", "0.0.0.0:5000", "app:app"]
```